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EXERCISE: LOG FILE ANALYSIS FOR INCIDENT DETECTION LAB

Aim

To analyze system log files and detect suspicious activities such as failed login attempts and identify suspicious IP addresses.

Objective

- Understand system log files
- Detect suspicious activities and incidents
- Analyze logs using Linux tools
- Identify attack patterns (failed logins, brute force, etc.)

Requirements

- Kali Linux / Ubuntu VM
- Basic Linux command knowledge
- Tools:
 - Kali Linux
 - journalctl
 - ssh
 - grep, awk, sort
 - grep
 - awk
 - less
 - cat
 - journalctl

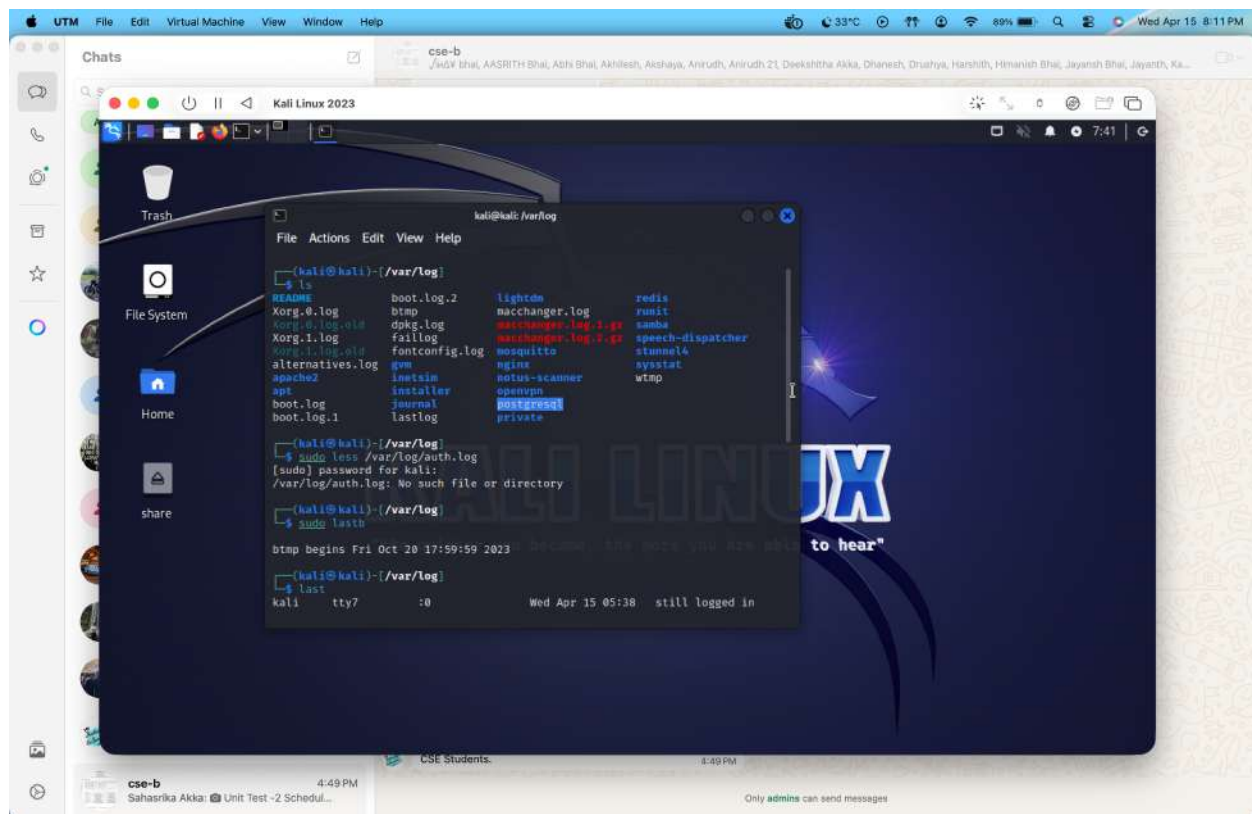
Understanding Log Files

Navigate to Log Directory

cd /var/log

ls

The **/var/log** directory contains system and application logs such as Apache, Nginx, and login records used for security analysis.



What Your Output Shows

From your `ls`:

- ✓ `dpkg.log` → package activity
- ✓ `apache2` → web server logs
- ✓ `nginx` → web logs
- ✓ `journal` → system logs (IMPORTANT)
- ✓ `wtmp`, `btm` → login records

Successful Logins (`wtmp`)

Command:

`last`

Capture:

- Login history

This displays successful login sessions. It helps identify legitimate and suspicious user access.

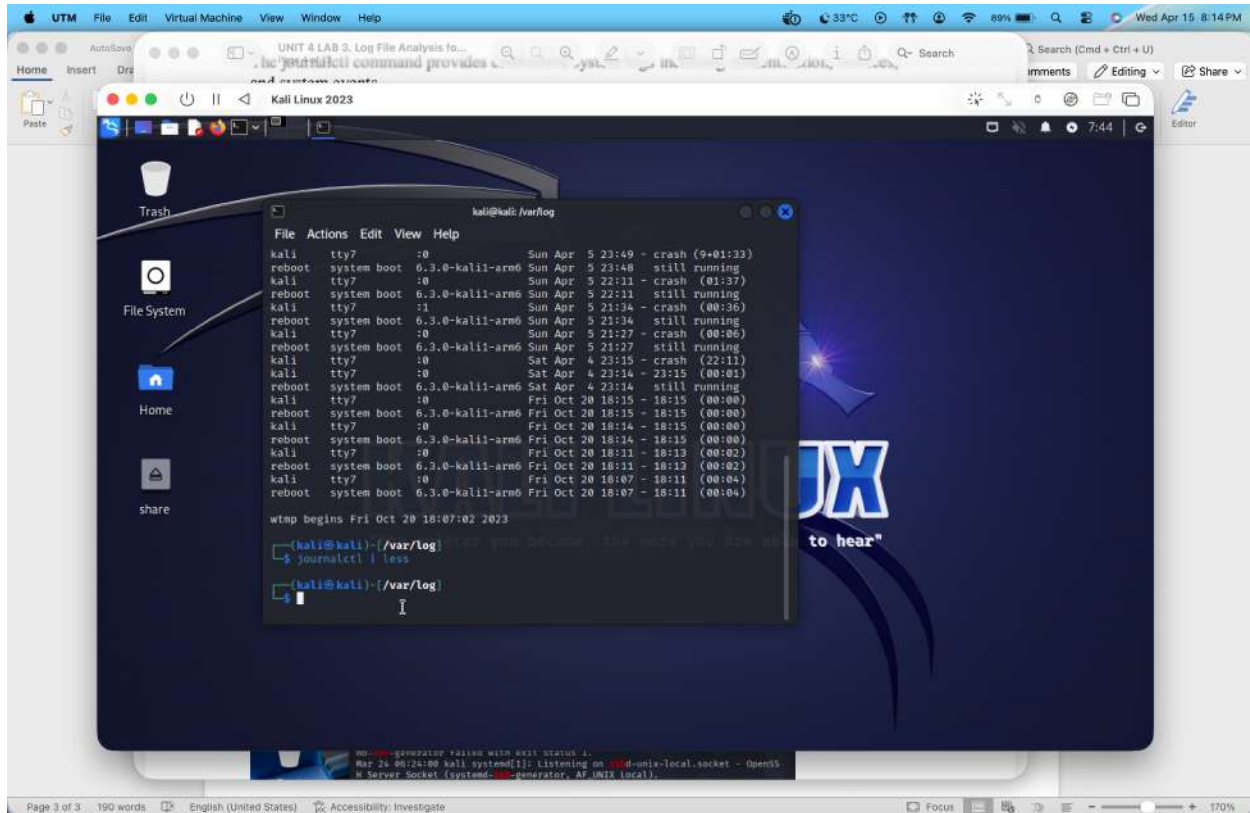
System Logs using journalctl

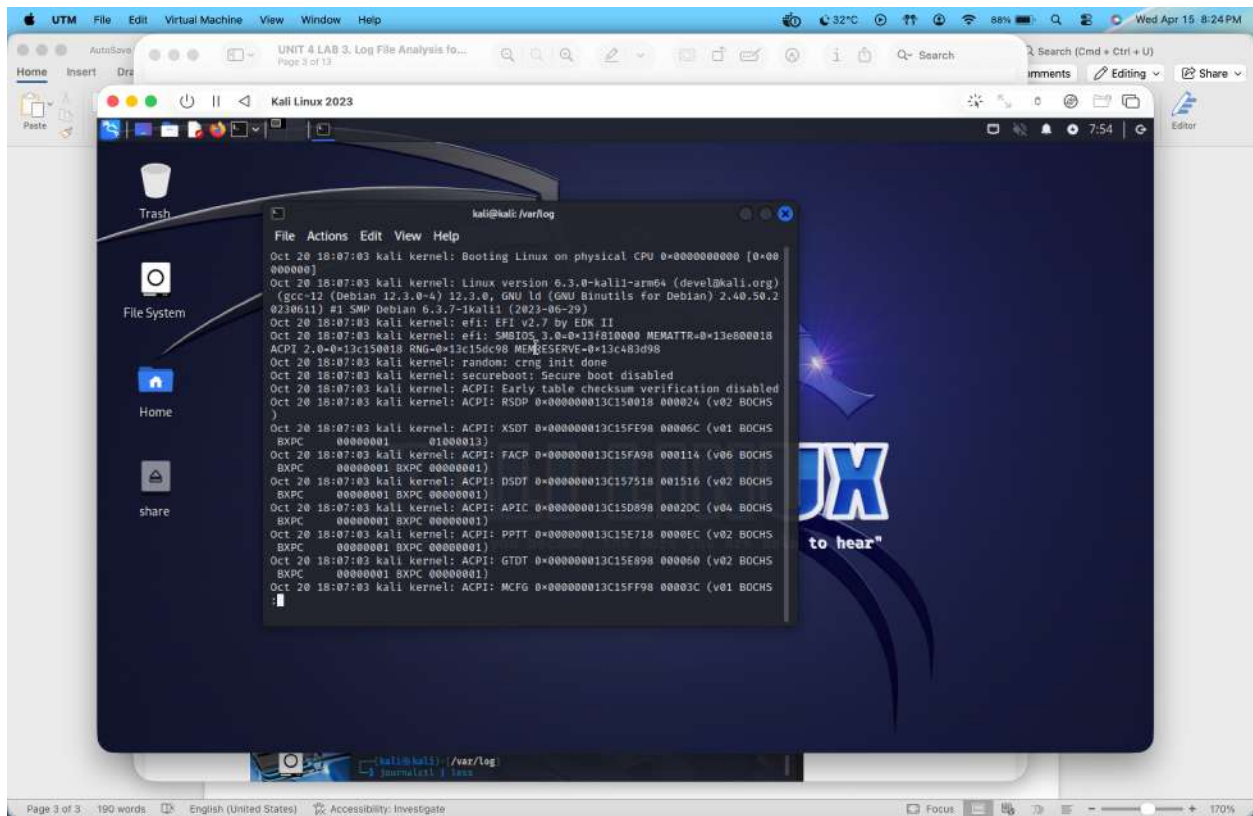
Command:

```
journalctl | less
```

Capture:

- Logs output

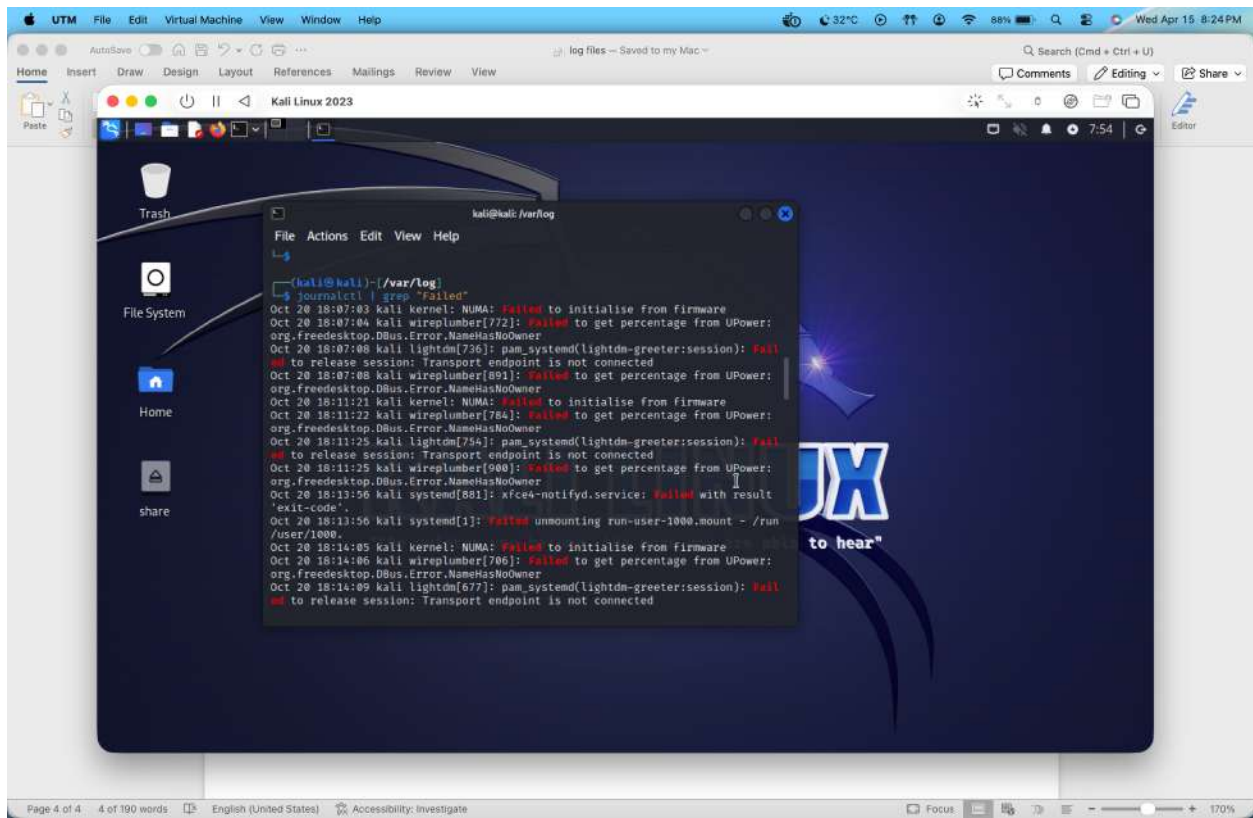




Failed SSH Attempts

Command:

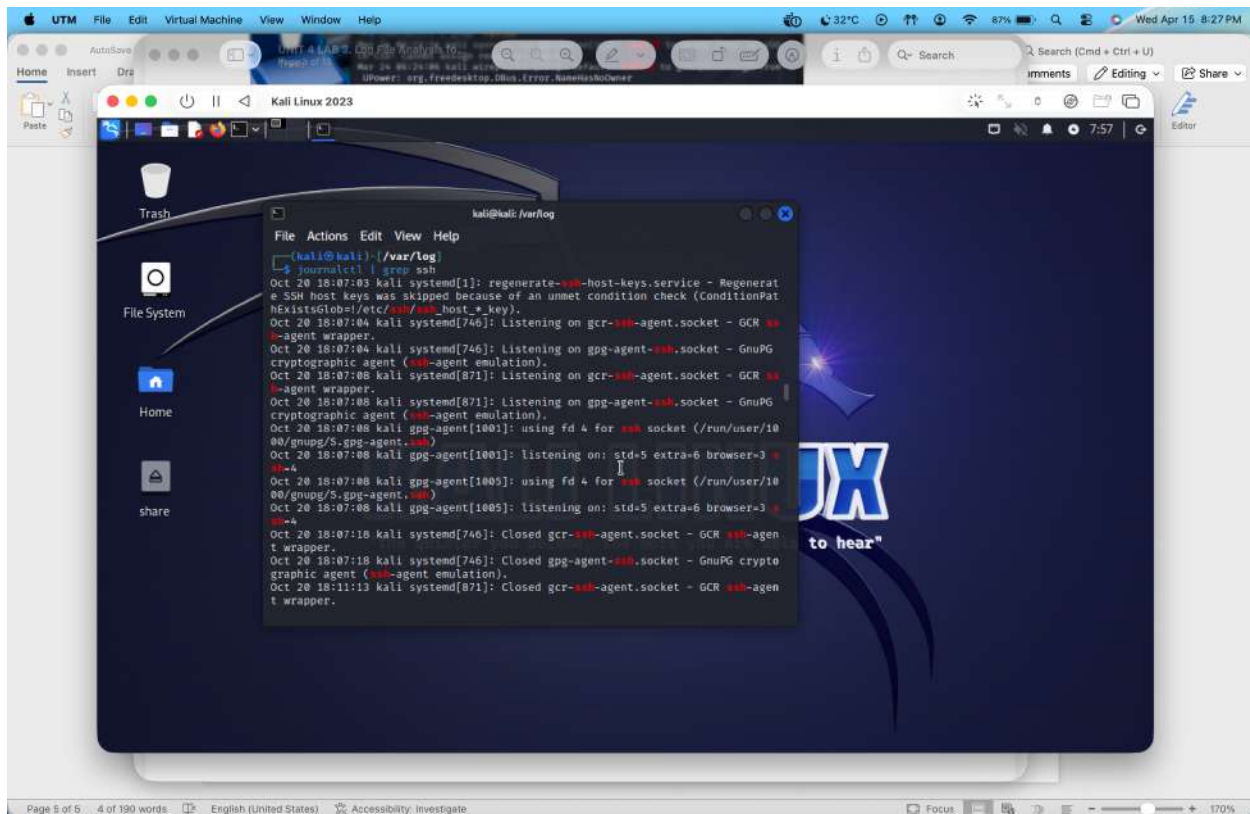
```
journalctl | grep "Failed"
```



SSH Login Activity

Command:

```
journalctl | grep ssh
```

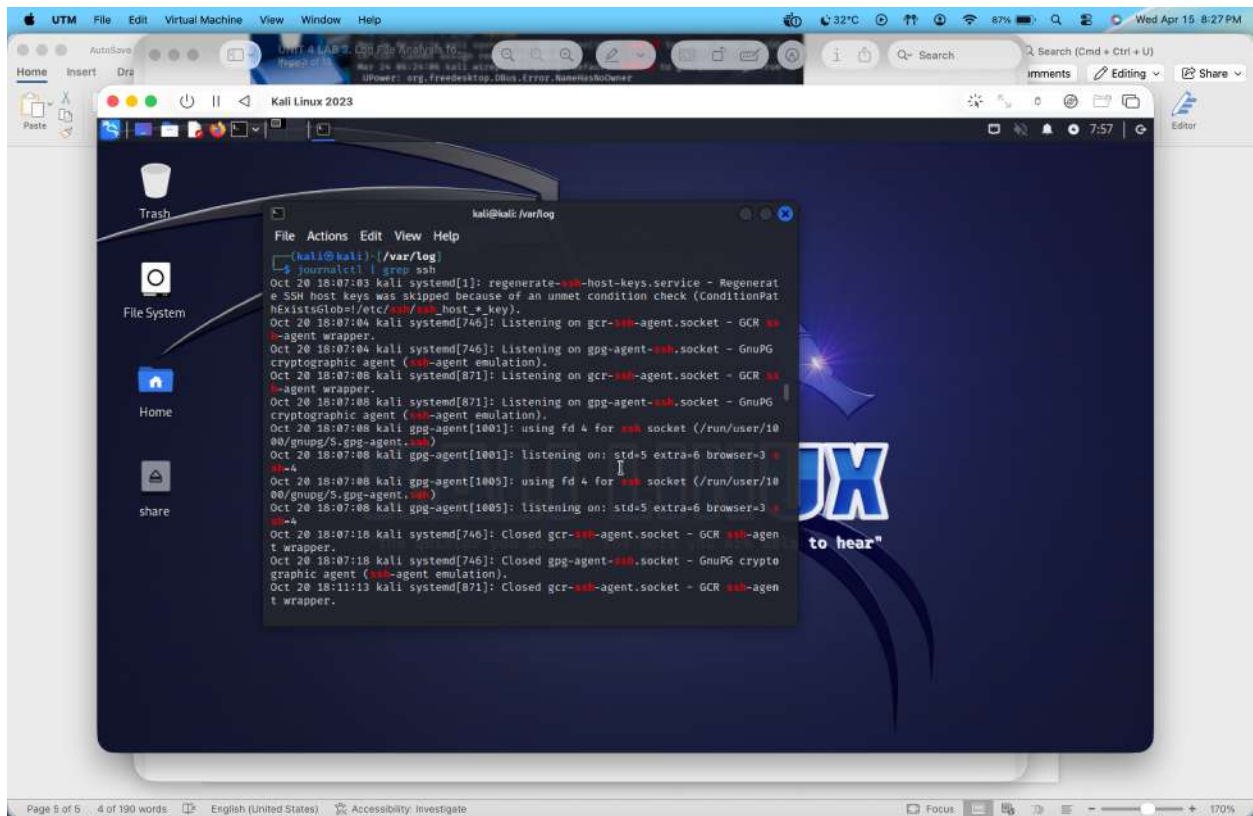


Explanation:

The journalctl command provides centralized system logs including authentication, services, and system events.

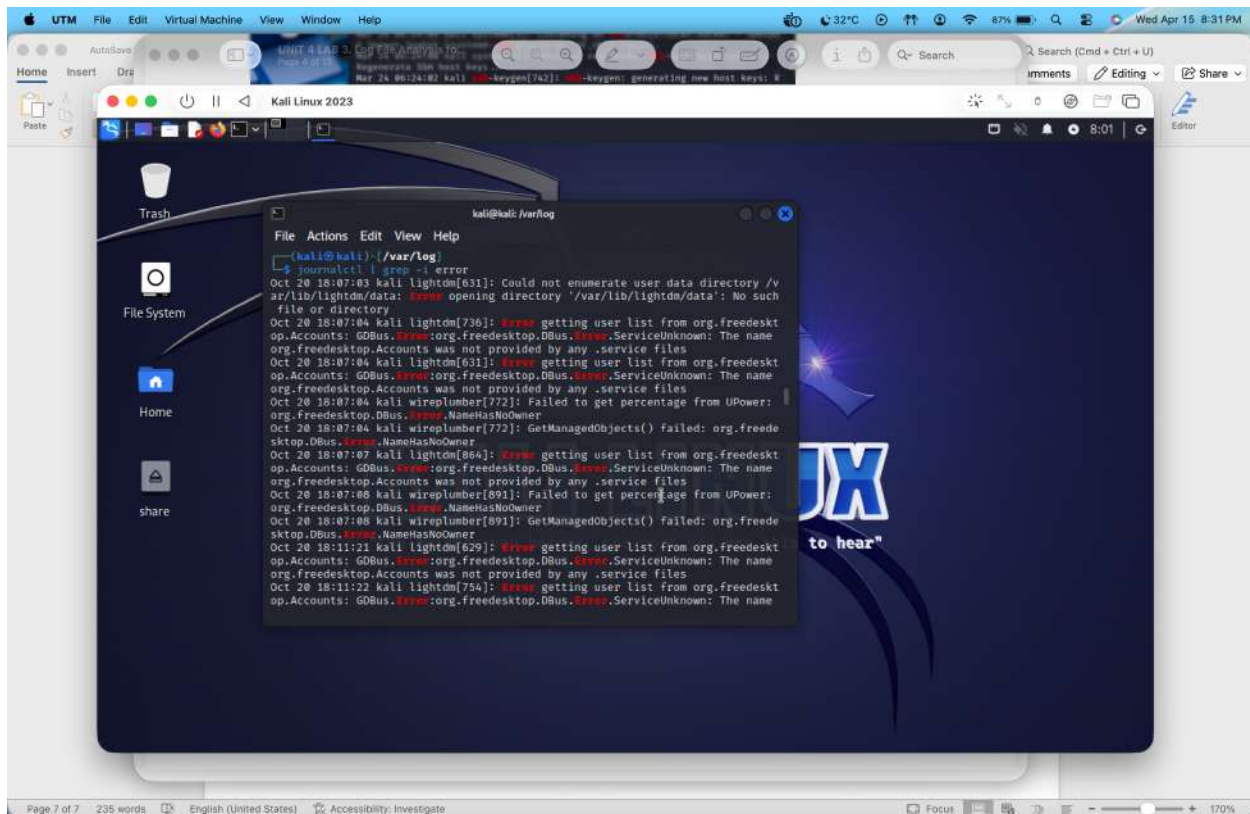
This screenshot shows failed authentication attempts, which may indicate unauthorized access attempts.

This screenshot also displays SSH-related logs, including login attempts and connections.



Error Detection

Command: `journalctl | grep -i error`



The above screenshot highlights system errors, which may indicate abnormal behavior or system issues.

Apache Log Analysis

Command:

```
cd /var/log/apache2
```

```
ls
```

Then:

```
sudo less access.log
```

Apache logs record web requests. These logs help identify suspicious web activity such as repeated requests or attacks.

Suspicious Requests

Command:

```
grep "404" /var/log/apache2/access.log
```

This shows failed web requests (404 errors), which may indicate scanning or probing attacks.

Package Activity Command:

```
less /var/log/dpkg.log
```

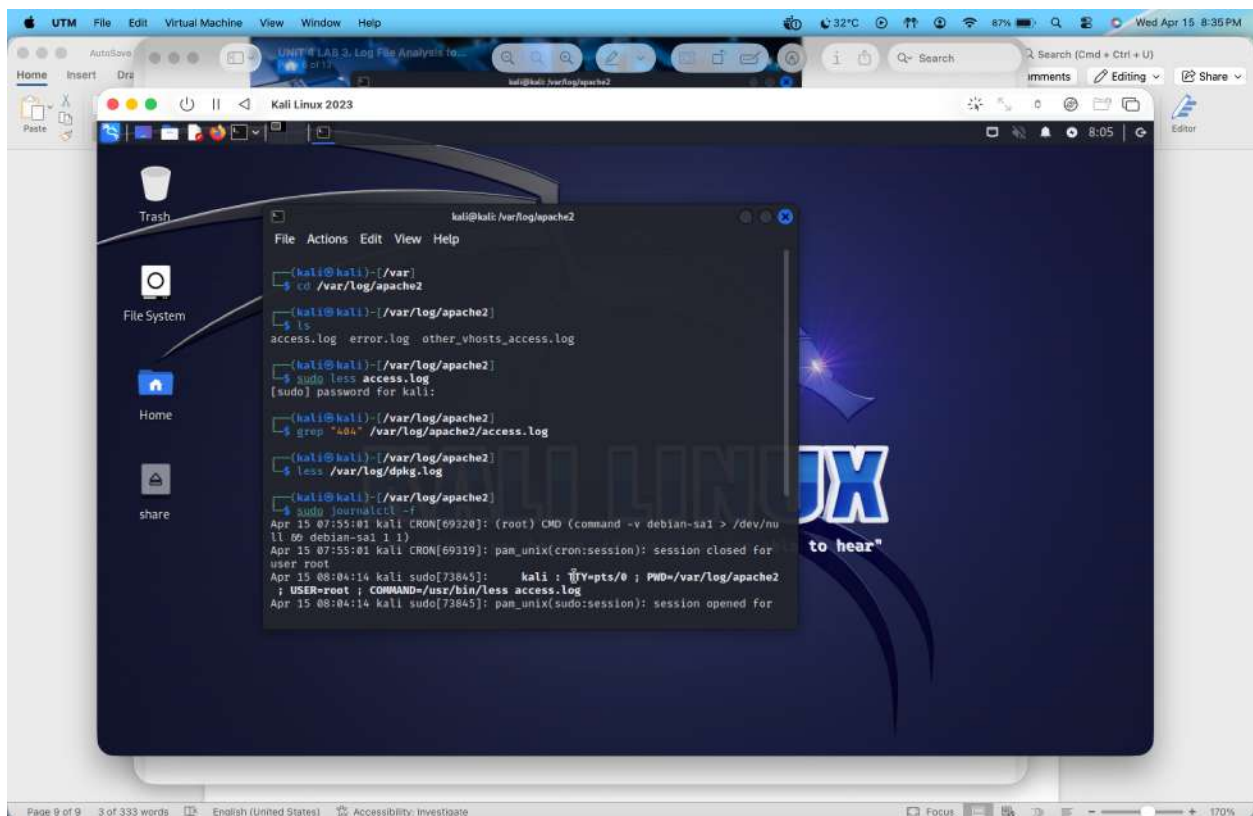
This shows software installation and update logs, useful for detecting unauthorized changes.

Real-Time Monitoring

Command:

```
sudo journalctl -f
```

This shows real-time system log monitoring, allowing detection of incidents as they occur.



QUICK CHECKLIST

Check	Command	Suspicious If
Failed logins	lastb	Many attempts
SSH activity	`journalctl	grep ssh`
Login history	last	Odd timing
Errors	`journalctl	grep error`
Web logs	grep 404	Many failed requests

STEP-BY-STEP ANALYSIS OF YOUR LOG

1. System Services (NORMAL)

Finished phpsessionclean.service

Finished systemd-tmpfiles-clean.service

- ✓ These are routine background tasks
- ✓ No issue

Regular system maintenance services were observed, indicating normal system operation.

2. CRON Jobs (NORMAL)

CRON: session opened for user root

(root) CMD (debian-sa1)

- ✓ Automated scheduled task
- ✓ Runs periodically

Scheduled cron jobs were observed, which are normal automated system processes.

3. Your Own SUDO Activity (NORMAL)

kali : COMMAND=/usr/bin/less access.log

kali : COMMAND=/usr/bin/journalctl -f

- ✓ This is YOU running commands
- ✓ Not suspicious

Sudo activity corresponds to legitimate administrative actions performed by the user.

4. XFCE / GUI Services (NORMAL)

xfconfd.service started

- ✓ Desktop environment service
- ✓ Normal

5. IMPORTANT PART

Repeated Kernel Messages

dndHGCM: Received message HOST_DND_FN_GH_REQ_PENDING
dnd: No guest source window These lines are:

Repeated MANY times
Coming from VirtualBox (VBoxDRM / DnD)

Is This Suspicious?

Technically:

- NOT an attack
- But abnormal repetitive behavior

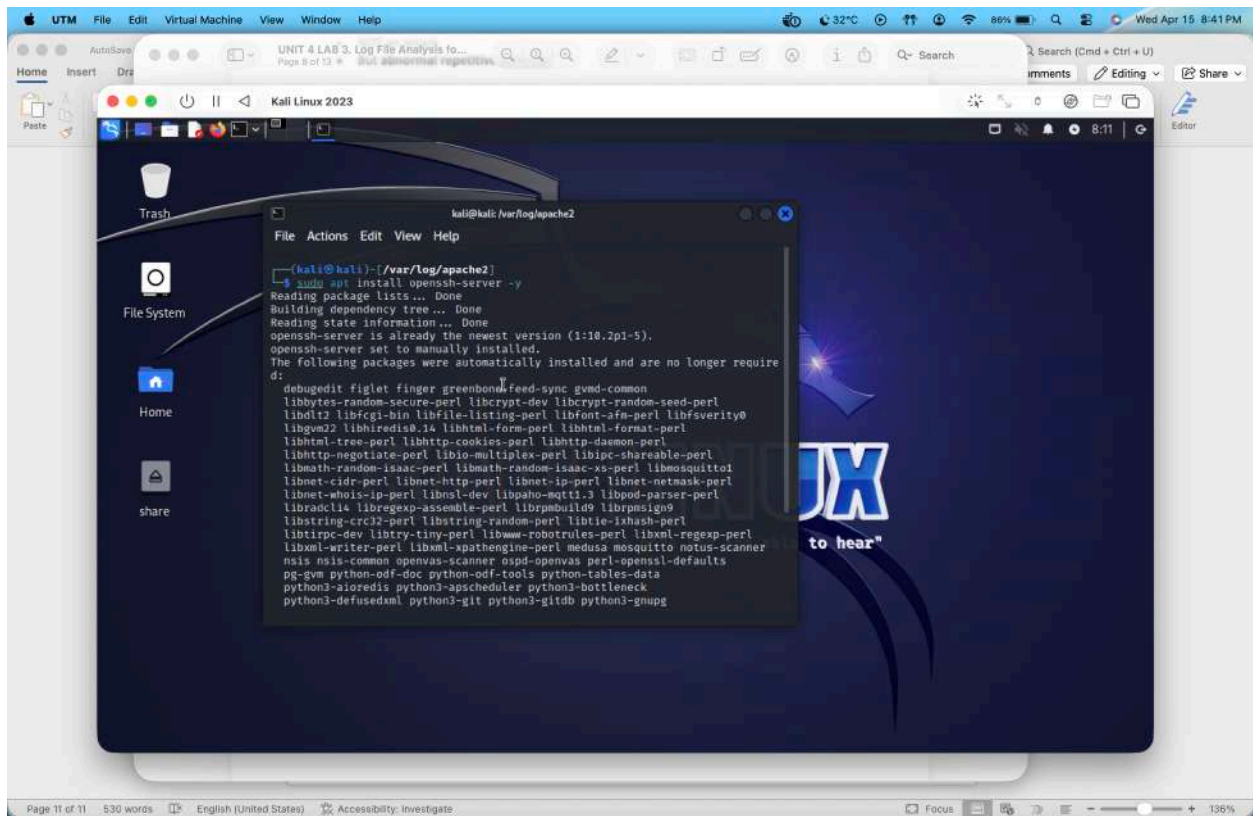
Why it matters:

- Repeated logs = unusual system activity
- Could indicate:
 - Misconfiguration
 - Resource issue
 - VirtualBox interaction problem

METHOD 1: SIMULATE FAILED SSH LOGIN

Step 1: Ensure SSH is Installed

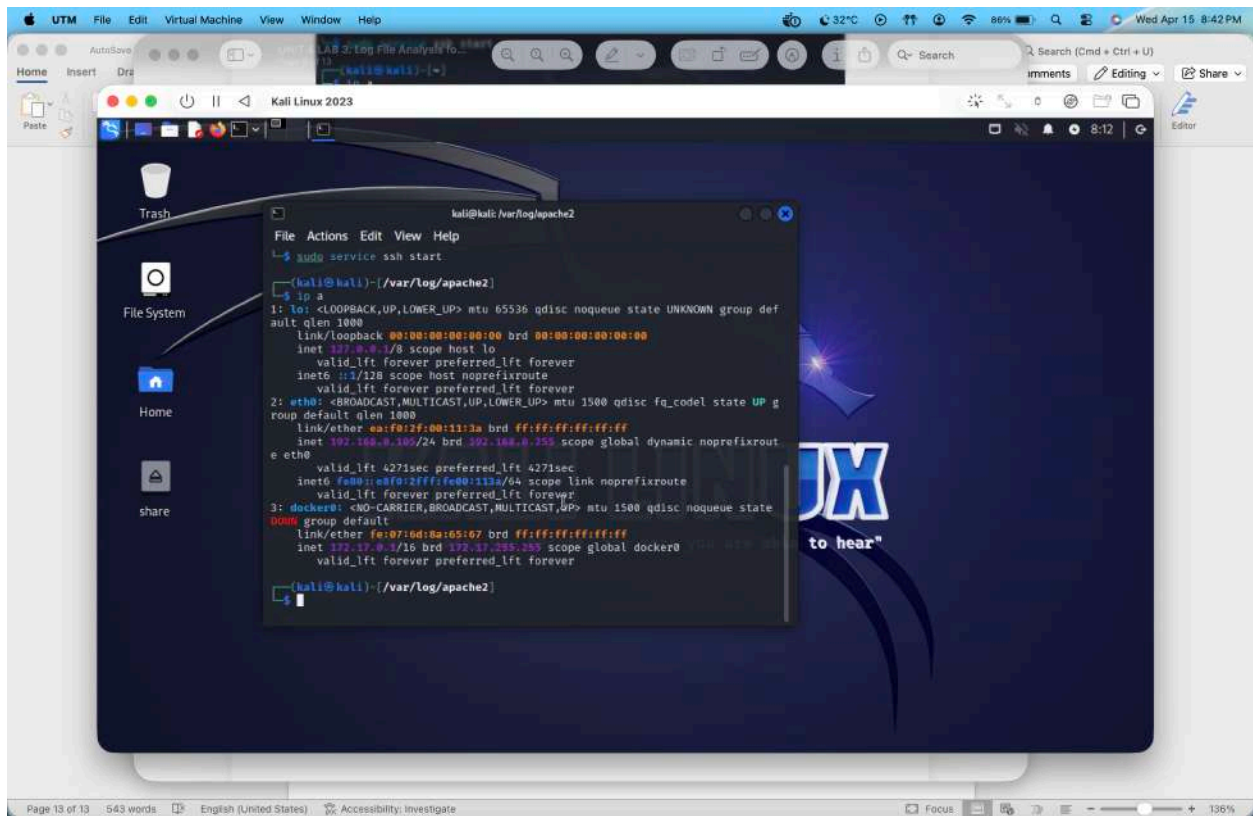
sudo apt install openssh-server -y



Step 2: Start SSH Service

sudo service ssh start

Look for something like: `inet 192.168.x.x`



Step 4: Attempt Wrong Login (GENERATE FAILURES)

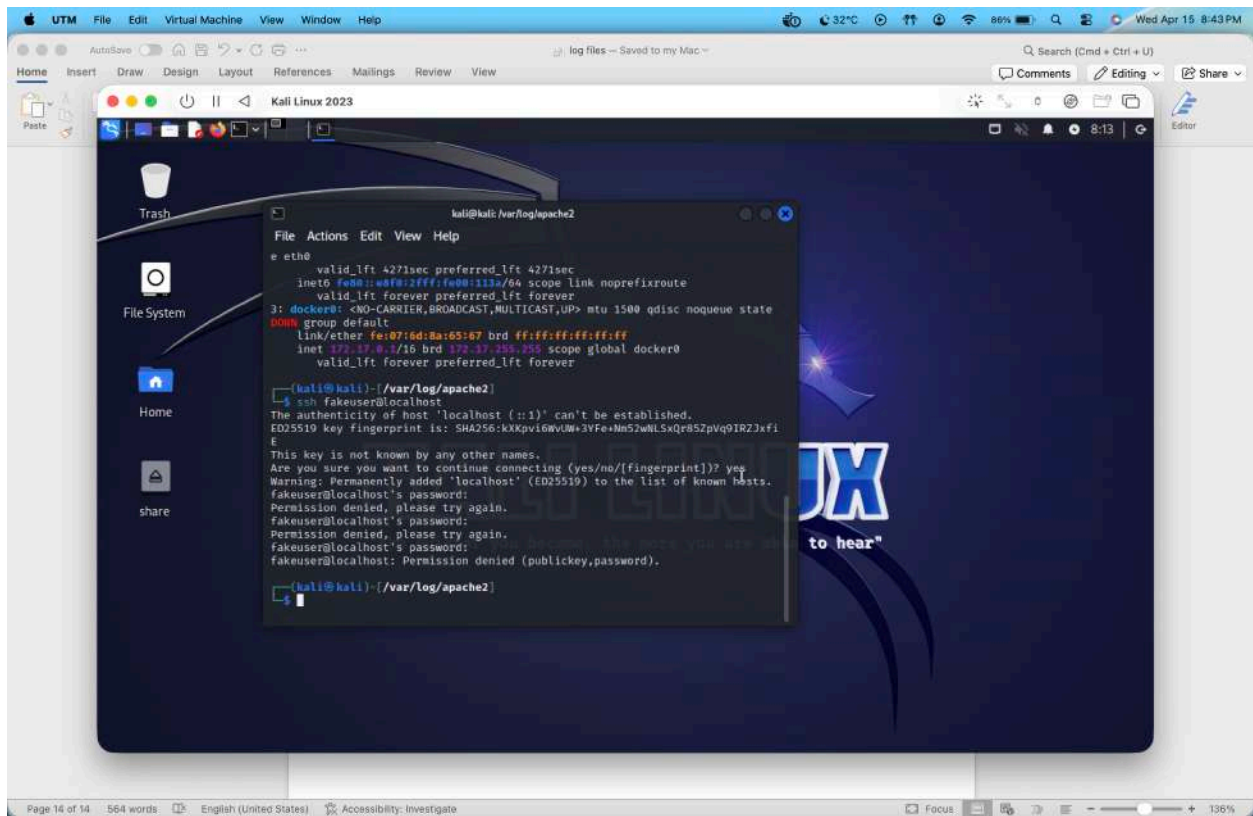
run:

```
ssh fakeuser@localhost
```

OR

```
ssh kali@localhost
```

Enter wrong password multiple times (5–10 times)



Now Analyze Logs

Step 5: Check Failed Logins

```
journalctl | grep "Failed password"
```

👉 You should now see:

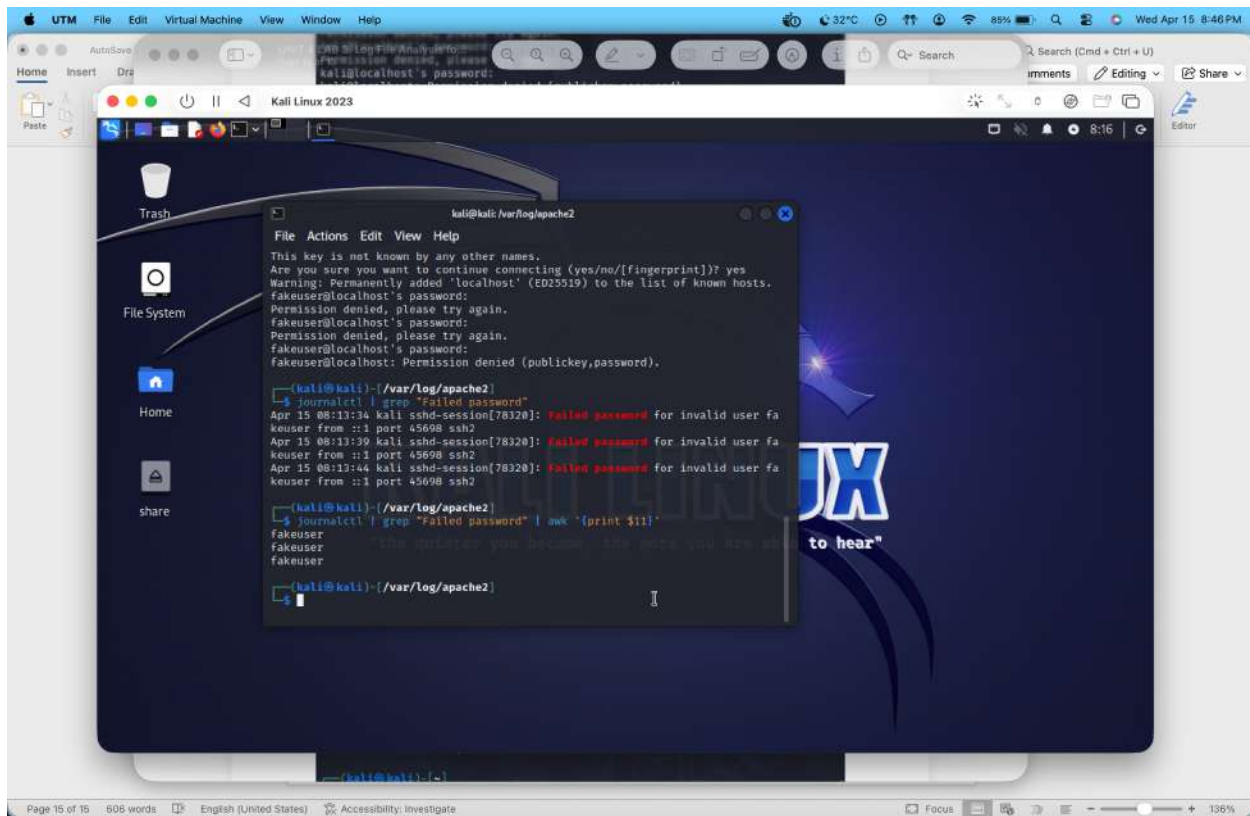
```
Failed password for kali from 127.0.0.1 port ...
```

Step 6: Extract Suspicious IP

```
journalctl | grep "Failed password" | awk '{print $11}'
```

Output:

```
127.0.0.1
```

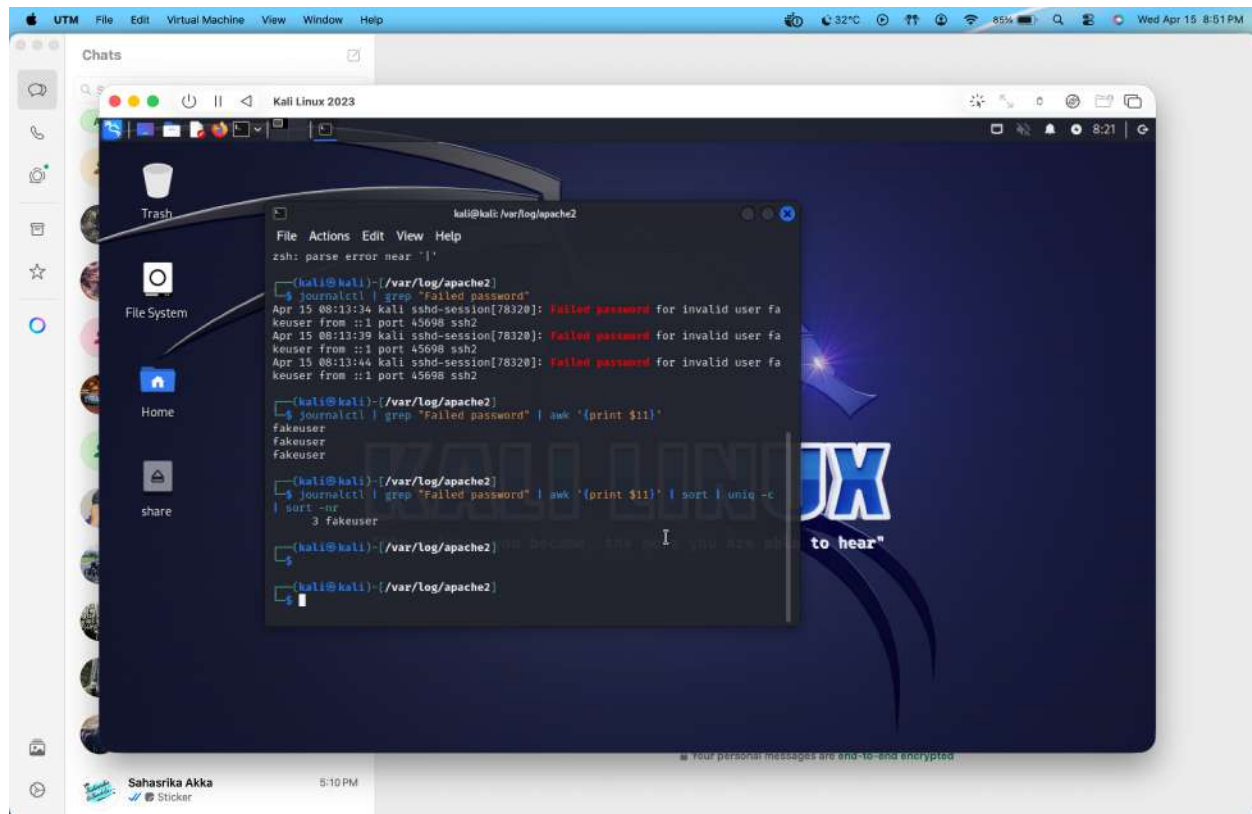



Step 7: Count Attempts Per IP

```
journalctl | grep "Failed password" | awk '{print $11}' | sort  
| uniq -c | sort -nr
```

Example:

```
10 127.0.0.1
```



INTERPRETATION (VERY IMPORTANT)

Multiple failed login attempts were detected from IP address 127.0.0.1, indicating a simulated brute force attack.

METHOD 2: Use lastb (Alternative)

```
sudo lastb
```

Shows failed login records

HOW TO IDENTIFY “SUSPICIOUS IP”

Simple rule:

Condition	Meaning
Same IP repeated many times	 Suspicious
Unknown IP	 Suspicious
High frequency attempts	 Brute force

Run:

```
journalctl --since "10 minutes ago"
```

Shows recent attack activity

